

# Overfitting And Underfitting (Bias And Variance)

- ① Training dataset
- ② Test dataset
- ③ Validation dataset

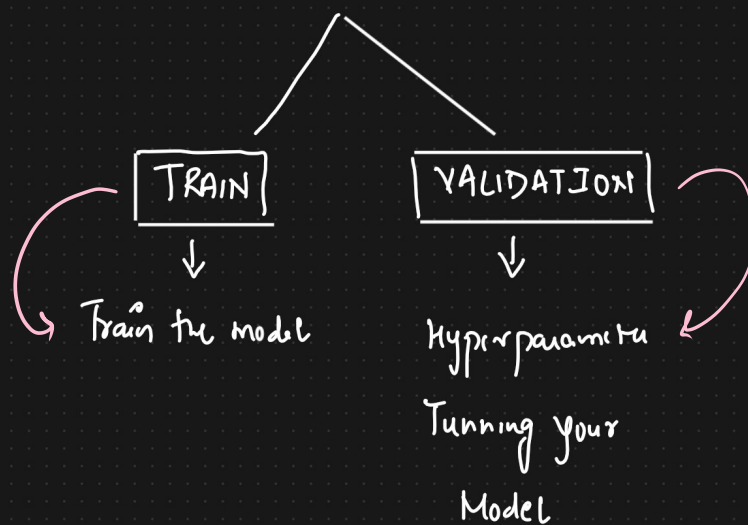
70%-30%

Dataset  
1000 datapoints

→ TRAINING DATASET (700) → Model To TRAIN  
→ TEST DATASET (300) → Testing Model

| Size of house | No. of bedrooms | Price |
|---------------|-----------------|-------|
| —             | —               | —     |
| —             | —               | —     |
| —             | —               | —     |
| —             | —               | —     |

## TRAINING DATASET



Now Let's Understand when we have underfitting & overfitting?

|                                                                            |                                                                                         |                                                                         |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <div> <div>DATA</div> <div> <div>TRAIN</div> <div>TEST</div> </div> </div> | <div> <div>(90%)</div> <div>Very Good Accuracy [Low Bias]</div> </div>                  | <div> <div>Very Good Accuracy (90%) [Low Bias]</div> </div>             |
|                                                                            | <div> <div>(85%)</div> <div>↑</div> <div>Very Good Accuracy [Low Variance]</div> </div> | <div> <div>↓</div> <div>Bad Accuracy (50%) [High Variance]</div> </div> |

Goal  
 Aim → Generalized Model

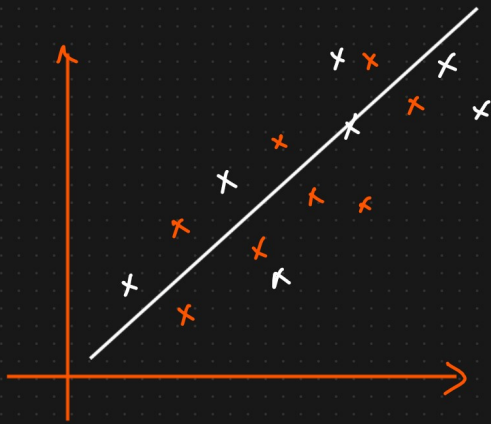
Model is Overfitting

TRAIN Model Accuracy is low [High Bias]

TEST Model Accuracy is low [High Variance]



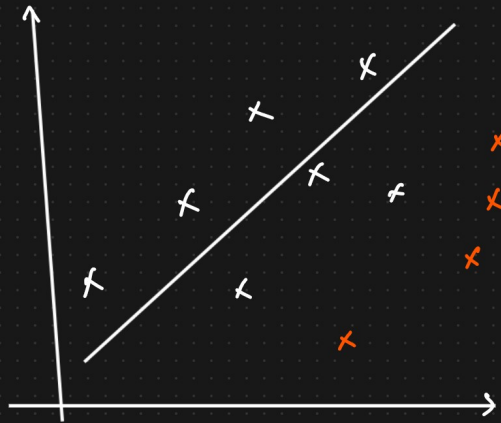
Model is Underfitting



Generalized Model



Low Bias, Low Variance



Overfitting

Low Bias, High Variance